

A
MAJOR PROJECT REPORT
ON
“College Management System”

Submitted in the partial fulfillment of the requirement for the
award of degree of

MASTER'S OF COMPUTER AND APPLICATIONS
IN
COMPUTER SCIENCE & ENGINEERING



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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

INDIRA GANDHI UNIVERSITY MEERPUR, REWARI

Session (2024- 2026)

INDIRA GANDHI UNIVERSITY MEERPUR REWARI

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE OF COMPLETION

This is to certify that the major project entitled "**College Management System**" carried out by **Ms.Himanshi**, student of MCA in Computer Science and Engineering 4th Semester, of Indira Gandhi University, Meerpur, Rewari is a satisfactory account of the bonafide work under my supervision, is recommended towards the end of her 4th semester of MCA (CSE).

Project Incharge

DECLARATION BY THE CANDIDATE

I, **Ms. Himanshi**, hereby declare that the major project work entitled is an "**College Management System**" authenticated work carried out by me at **Yhills Edutech Pvt. Ltd** for the partial fulfillment of the award of the degree of MCA, Indira Gandhi University, Meerpur, Rewari and this work has not been submitted for similar purpose anywhere else.

Date:

Place: Rewari

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Roll No. : 211011107003

MCA 4th SEM

ACKNOWLEDGEMENT

First of all I will thank to our mam,“ Dr. Sangeeta ”, who give me the valuable suggestion for my minor project.

The success of this final report is the outcome of Guidance and Valuable suggestions provided by the all concerned without which the report could not fide on the right back.

I express my sense of deep gratitude to Assistant Professor “Dr. Sangeeta” for inclusions and timely suggestions in the preparation of this final report.

Himanshi

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MCA 4th SEM

Abstract

This project report presents the design and implementation of a College Management System using Visual Basic as the front-end and Oracle as the back-end. The objective of this project is to develop a comprehensive and user-friendly application that automates and streamlines various academic and administrative tasks in a college setting.

The system is designed to facilitate the effective management of student records, faculty information, attendance tracking, examination details, fee payment, and timetable scheduling. It ensures secure access and role-based privileges for different types of users—administrators, faculty members, and students—thus maintaining data confidentiality and integrity.

The report details the planning, analysis, design, and theoretical implementation of the system, excluding the source code. It includes system design diagrams, data flow diagrams, entity-relationship models, and testing methodologies. A detailed feasibility study is also included to ensure the technical, operational, and economic viability of the system.

Through this project, we aim to demonstrate the potential of integrating database systems with application interfaces to build efficient and robust management solutions for educational institutions.

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INTRODUCTION

Introduction to Project

In the modern educational landscape, managing institutional operations through digital systems has become a necessity rather than a luxury. The College Management System is designed to digitize and streamline various academic and administrative tasks of a college. Developed using Visual Basic as the front-end and Oracle as the back-end database, this system offers a comprehensive solution to manage students, faculty, staff, courses, examination schedules, attendance, fee transactions, and other essential operations.

This system reduces manual intervention and ensures that critical information is available in real-time to all stakeholders—students, faculty, and administrators. It replaces traditional paper-based systems and excels in providing accuracy, speed, security, and data integrity.

The College Management System is intended to work as an integrated platform where multiple user roles interact. For example, an admin can manage user accounts, schedules, courses, and examinations, while faculty members can manage attendance and grades, and students can access their profiles, fee records, and results.

This report provides a comprehensive overview of the project including its objectives, design methodology, system components, testing procedures, and potential for future enhancements. Each component of the system has been carefully analyzed and developed to ensure maximum efficiency and effectiveness in managing institutional processes.

Problem Statement

Educational institutions, particularly colleges, often face challenges in efficiently managing their operations when using manual or semi-digital systems. Traditional methods involve maintaining bulky records, ledgers, and registers, which are not only time-consuming to update and retrieve but also prone to human errors and data redundancy.

Some common problems faced in the manual system include:

- Difficulty in managing and retrieving student records, especially as the number of students increases annually.
- Time-consuming fee processing and record-keeping, which often leads to delays and inconsistencies.
- Inaccurate attendance tracking, which can affect internal assessments and student evaluations.
- Complex scheduling of classes, exams, and faculty assignments, leading to conflicts and inefficiencies.
- Lack of centralized control and real-time accessibility for different stakeholders.

These limitations create operational bottlenecks and diminish the overall efficiency of the academic institution. In addition, manual processes do not offer transparency, audit trails, or data security features which are essential in today's educational environment.

Hence, there is a pressing need for a centralized College Management System that can automate and integrate all these functions into a single platform to ensure accuracy, accountability, and real-time access to information.

Objectives of the Project

The primary goal of the College Management System is to automate and streamline core academic and administrative functions within a college, ensuring efficiency, accuracy, and transparency. The specific objectives of this project include:

9.1 Centralized Information Management

To create a centralized database where all student, faculty, course, attendance, fee, and examination data are stored securely and can be accessed based on user roles.

9.2 Efficient Student Lifecycle Management

To manage the complete lifecycle of a student—from admission, attendance tracking, performance monitoring, to fee submission and final result generation.

9.3 Automation of Routine Tasks

To reduce manual intervention in repetitive tasks such as attendance marking, report generation, and timetable scheduling, thereby saving time and resources.

9.4 Role-Based Access Control

To ensure that different users (admin, faculty, student) can access only the relevant data and functionalities according to their role in the institution.

9.5 Accurate and Real-Time Reporting

To generate accurate reports and analytics (e.g., attendance summary, student performance, fee collection) in real time for improved decision-making.

9.6 Enhanced Communication

To enable better communication between students, faculty, and administrative staff through timely notifications and information updates.

9.7 Secure and Reliable System

To develop a system with built-in security features to prevent unauthorized access, data loss, or modification.

Scope of the Project

The College Management System has been developed to cover a wide range of institutional processes and is scalable to meet the needs of small to large-sized colleges. The scope of the project encompasses:

10.1 User Management

- Admin can create and manage user accounts for students, faculty, and other staff.
- Different access rights and dashboards are provided based on the user role.

10.2 Student Information Management

- Storage and management of student personal data, academic records, attendance, and fee details.
- Students can view their profiles, timetables, attendance status, and results.

10.3 Faculty and Staff Module

- Faculty members can view student lists, mark attendance, enter exam marks, and access schedules.
- Faculties can also update course content and communicate with students if needed.

10.4 Attendance Management

- Daily attendance can be recorded by faculty members.
- The system generates consolidated attendance reports and alerts for short attendance.

10.5 Fee Management

- Records fee structures, due dates, and payment statuses.
- Generates receipts and tracks due payments, with the facility to generate fee reports.

10.6 Examination Module

- Management of internal and external exams, mark entry, and result computation.
- Generation of grade reports and result sheets.

10.7 Timetable Scheduling

- Admin can create class schedules while avoiding conflicts for classrooms and faculty.
- Timetables are visible to both students and faculty.

10.8 Report Generation

- Generates various reports such as student performance, fee status, attendance summaries, and exam results.

10.9 Data Security and Backup

- The system uses Oracle DB to ensure data consistency and integrity.
- Regular backup mechanisms are integrated to protect against data loss.

10.10 Future Integration Possibilities

- SMS/Email notification services.
- Biometric integration for attendance.
- Online portal for parent access.

Preliminary System Analysis

Preliminary System Analysis

i) Identification of need:

Identification of need of a system is very important for developing new system or redefining the existing system as the accuracy of any system design depends upon the accuracy in finding out the actual need and the actual problem and the requirement of the user.

The basic reason behind this step is to investigate the actual problem faced by the user in the present manual system. This will give an idea about the future requirement i.e. whether the user want to create a new system or want to modify the current system.

Before developing this project we have identified the various problems faced at the box office. At present the system is working manually. Hence the computerized system is developed to meet the requirement of the user.

ii) Preliminary Investigation:

The basic input of any activity is data that would be later processed to obtain the desired information. In order to get fruitful information, it is necessary to collect related data. With the combination of various data we can get many types of information for the smooth functioning of any activity.

The paper work that has been used in any box office maintain the record of various booked tickets is very time consuming so it is very necessary to keep track of assigned tickets.

To meet all this requirement, a system is needed which will keep record of all the activities carried out at the box office in a well designed manner.

iii)Flaws in Present System:

The existing system is manual and has many flaws. The system did fulfill the requirement of the earlier time but today with over population and the bulky records the system has aged off.

- Less Accurate
- Less efficient
- Possibility in creating mistake in calculations
- Time consuming process.
- Bulk of paper work.
- The most important factor is that time and speed greatly affects during working in manual system

iv) Need of New System:

- This system reduces paper work.
- This system reduces errors.
- It is easy to access present data.
- Direct access by operator to seat booking information of the client.
- Systematic storage of record.

Only a single person can operate this system. It also provides proprietor best tools to access all the information about his business to work on it. The application deals with all those small aspects of business. We decided to create this entire project in VB6.0 because it provides graphical interface, user-friendly, easy to understand. Coding, compilation and debugging can be easily done and errors can be found more accurately and effectively. Access is used as database since it is very easy to handle and operate. In access it is also easy to access data into front end.

The logic behind rest of the operation is varying, simple to develop, as the desired record and searching of records is conventional. This project helps client to manage its various transactions and information related to its clients and management system.

v)Feasibility Study:

- Feasibility of a system means whether the system is practically possible to build or not. It also evaluates the benefits of the new system.
- Feasibility is evaluated from developer's and Students's point of view.
- The software/system which is to analyze in sufficient detail and it is to be developed is technically, operationally and economically feasible.

Technical Feasibility

Technical feasibility centers on the existing computer system i.e. the availability of the required hardware, software and the operating system. This project is developed in visual basics 6.0 which can be run on any system with the required configuration.

Operational Feasibility

The proposal system/software is user-friendly in nature. This system will be helpful for the user to perform all necessary operations efficiently and effectively. The Graphical User Interface is self-explanatory; as a result there is no need of providing any kind of training to the user about the software provided. Demonstration of the system/software once to the user will definitely help the user update the software efficiently.

Economic Feasibility

The cost of installing this project is not high. Once installed, the project gives long-term benefit. The cost of hardware and software that is required to build the system is very much within the reach of the organization. So, it is possible to go ahead with the proposed system from economic point of view

vi). Project Category:**ABOUT RDBMS**

RDBMS is the presentation of the data. It provides a set of more or less flexible and sophisticated tools for handling information. These tools support the following tasks.

- Defining a database

- Querying the database
- Adding editing or deleting data from the database
- Modifying the structure of the database
- Securing the data from public access
- Communicating within the networks
- Exporting and importing data

Because it gives so much control over the data, RDBMS can also serve as the foundation for products that generate applications and extract data .There are three major roles involved in the development, implementation or the using of the RDBMS.

They are :

- The database administrator who has a special set of privileges that give him a complete control over the data and includes such functions as creating primary database structures, modifying them, backing up and restoring them.
- The system developers who can create application ranging from simple data entry forms to comprehensive multi user application.
- The end user who must query the system and generate the output.

A good RDBMS and a well –designed Relational database give you some important benefits.

- Data integrity and consistency maintained and enforce by the RDBMS.
- Redundant data eliminated or kept to a practical minimum.
- Data retrieved by the unique key.
- Relationships expressed through matching keys.
- Physical organization of the data managed by RDBMS.
- Optimization of storage and database operation execution times.
- Concurrency: database users don't corrupt each other's work.

GUI (GRAPHICAL USER INTERFACE)

The GUI provides picture – oriented interface. Instead of typing command, it provides menus and icons (small representation pictorial images) which the programmer can select by pointing to with a mouse and then clicking one or its buttons. A GUI is software; WINDOWS present programs, procedures, files, commands, etc. as graphical symbols. The user interacts with the computer by manipulating the graphical symbols and menu instead of typing commands. WINDOWS is not an operating system. It can run under DOS. In DOS operating system the programmer has to type commands, when Windows -3 versions were introduced in 1990. Its later version is: WINDOWS -3.1, WINDOWS-3011 AND WINDOWS for Workgroups. Today the operating system WINDOWS -95 and WINDOWS -98 include WINDOWS graphical interface for their operation.

Software and Hardware Requirement Specification

SOFTWARE SPECIFICATIONS

Language: PHP

➤ Database: Ms-Access

➤ Operating System: Windows XP Professional, windows 7, windows 8.
(Resolution – 1024 by 768 pixels).

➤ Server : Visual Basic .NET

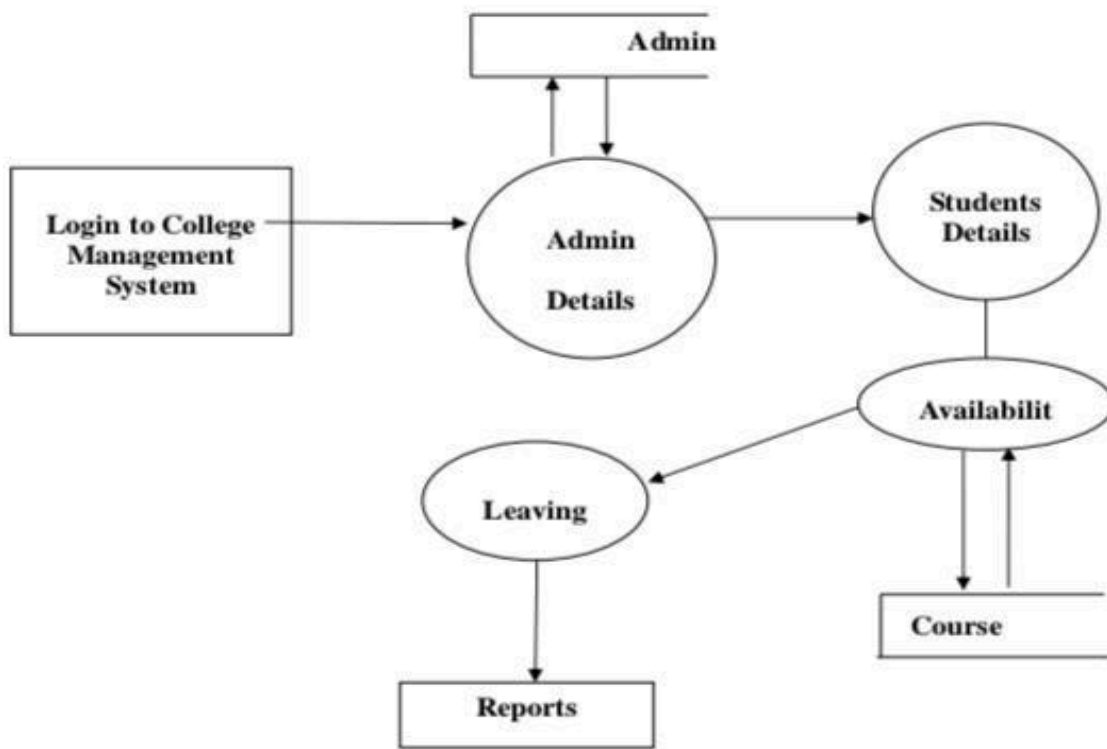
HARDWARE SPECIFICATIONS

Computer having: -

- | | |
|--------------------|----------------------|
| ▪ Processor | ▪ Monitor |
| Intel[R] Pentium 4 | 14" or above color |
| above | monitor |
| ▪ RAM | ▪ Printer |
| 512 MB of RAM | Black & White/ Color |
| ▪ Hard Disk | ▪ Keyboard & Mouse |
| 10GB or above | PS2 / USB |

Detailed System Analysis

- **Data Flow Diagram**



Data Flow Diagrams (DFD)

Data Flow Diagrams represent the flow of data in the system. It shows the inputs, processes, and outputs in a logical order.

Level 0 DFD (Context Level):

Shows the system as a single process.

External entities: Admin, Faculty, Student.

Data Stores: Student Info, Faculty Info, Attendance, Fees, Exam.

Data Flows: Login data, Registration data, Attendance status, Fee records, Marks, Timetable.

Level 1 DFD:

Breaks the main system into sub-processes:

1. Manage Student Information
2. Handle Attendance
3. Manage Exams and Results
4. Maintain Fee Records
5. Generate Timetable
6. Manage User Accounts

Each process interacts with external users and internal data stores.

• Number of Modules

1. Forms

- Admin
- Student
- Course
- Teacher
- Leaving

• Data Structure and Tables

TABLE: Student Table

Sr. No.	Field Name	Type	Size	Description
1	ID	Number	10	Student ID
2	Name	Text	20	Student Name
3	Quali	Text	35	Qualificatio n
4	Dob	Text	36	Date Of Birth
5	Add	Text	37	Address
6	City	Text	38	City
7	Mno	Number	35	Mobile No
8	Photo	Text	10	Photo

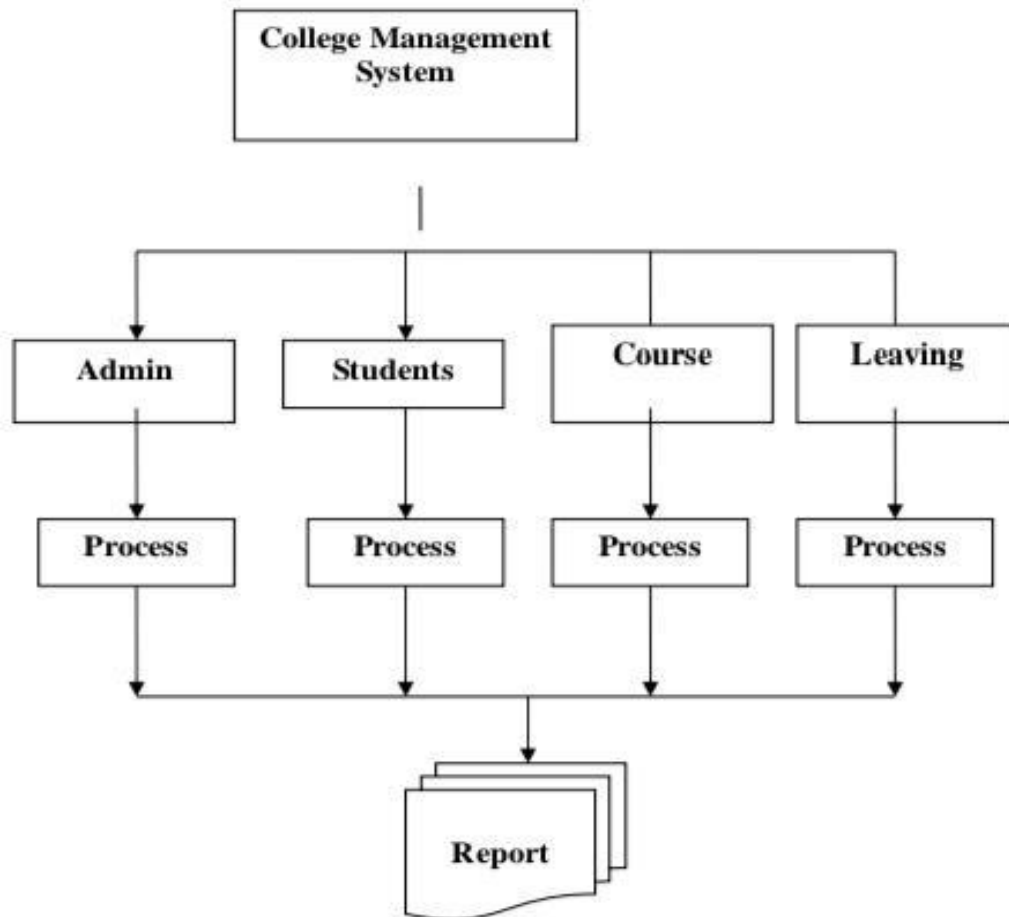
- **Table : Admin Table**

Sr. No.	Field Name	Type	size	Description
1	ID	Number	15	Course ID
2	Name	Text	20	Course Name
3	Add	Text	30	Address
4	City	Text	10	City
5	Mno	Number	10	Mobile No

- Table: Course Table

Sr. No.	Field Name	Type	size	Description
1	ID	Number	12	ID
2	Cname	Text	15	Course Name
3	Class	Text	25	Class name
4	Dur	Text	25	Duration
5	Fees	Text	25	Fees

- System Chart



• Entity Relationship Diagram

ER diagrams represent the relationships among entities in the database.

- Entities:
- Student
- Faculty

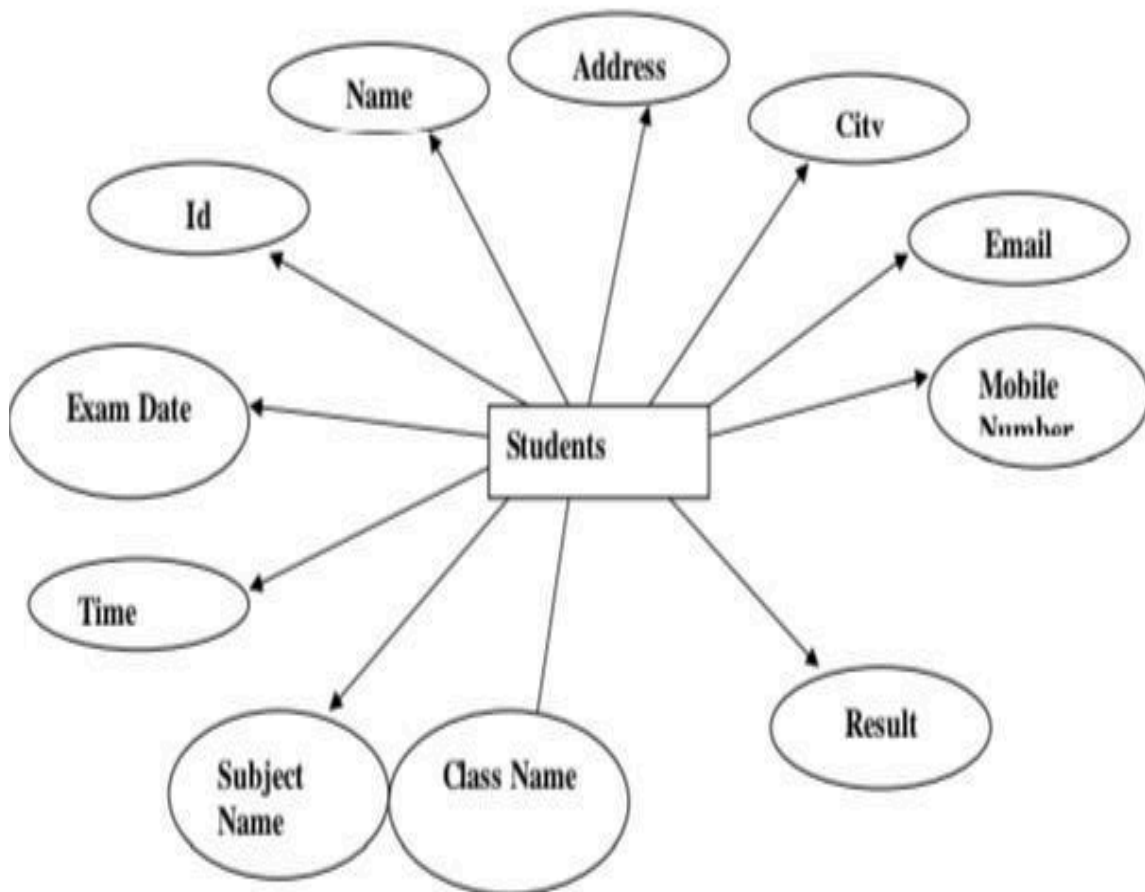
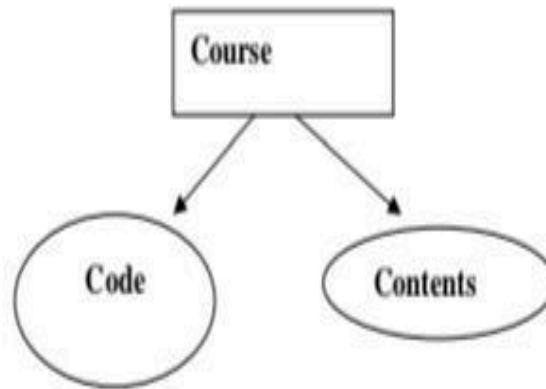
- Subject
- Course
- Attendance
- Exam
- Fees
- Timetable

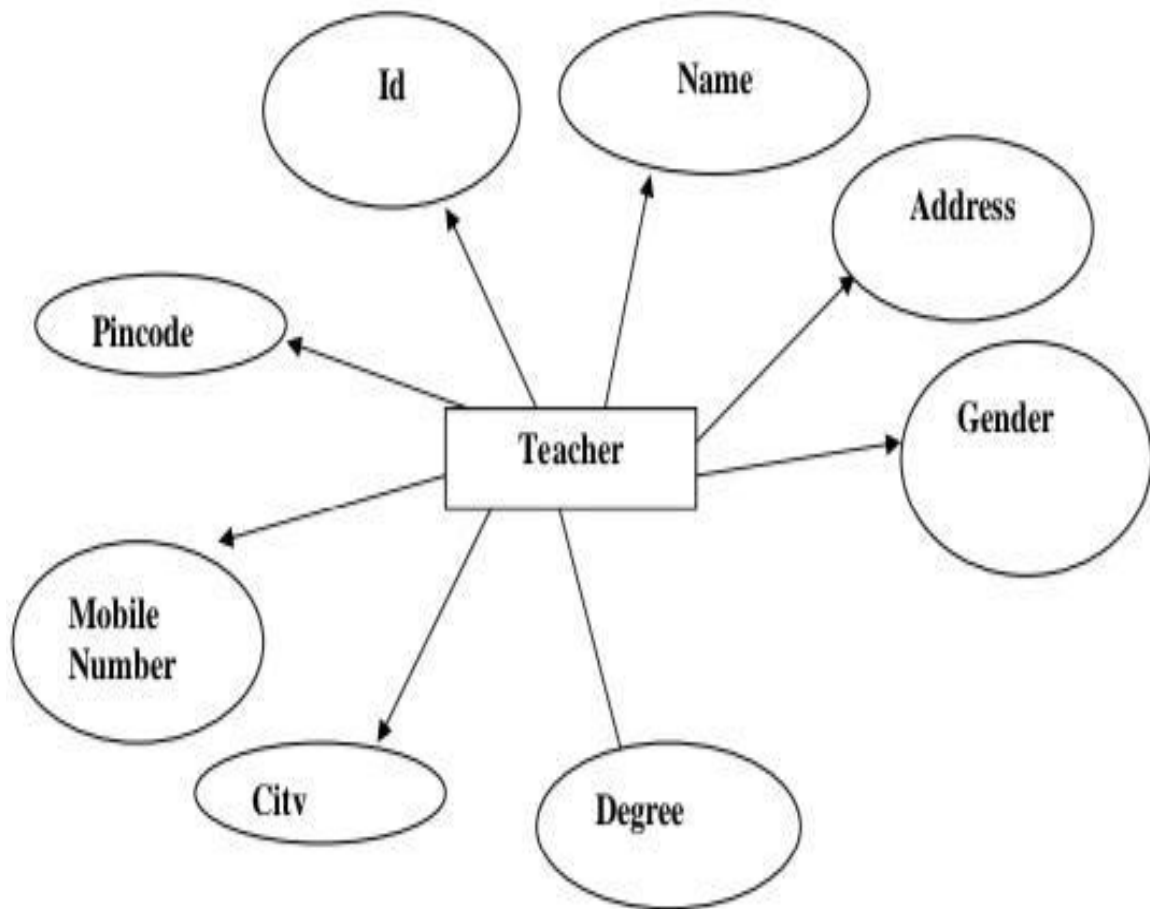
Attributes Example:

- Student: Student_ID (PK), Name, DOB, Course_ID (FK), Contact, Email
- Faculty: Faculty_ID (PK), Name, Department, Contact, Email
- Exam: Exam_ID (PK), Student_ID (FK), Subject_ID (FK), Marks

Relationships:

- One Course has many Students
- One Faculty teaches many Subjects
- One Student has many Exam entries
- One Student can make many Fee payments





Use Case Diagrams

Use case diagrams describe the system's functionality from the user's point of view.

Actors:

- Admin
- Faculty
- Student

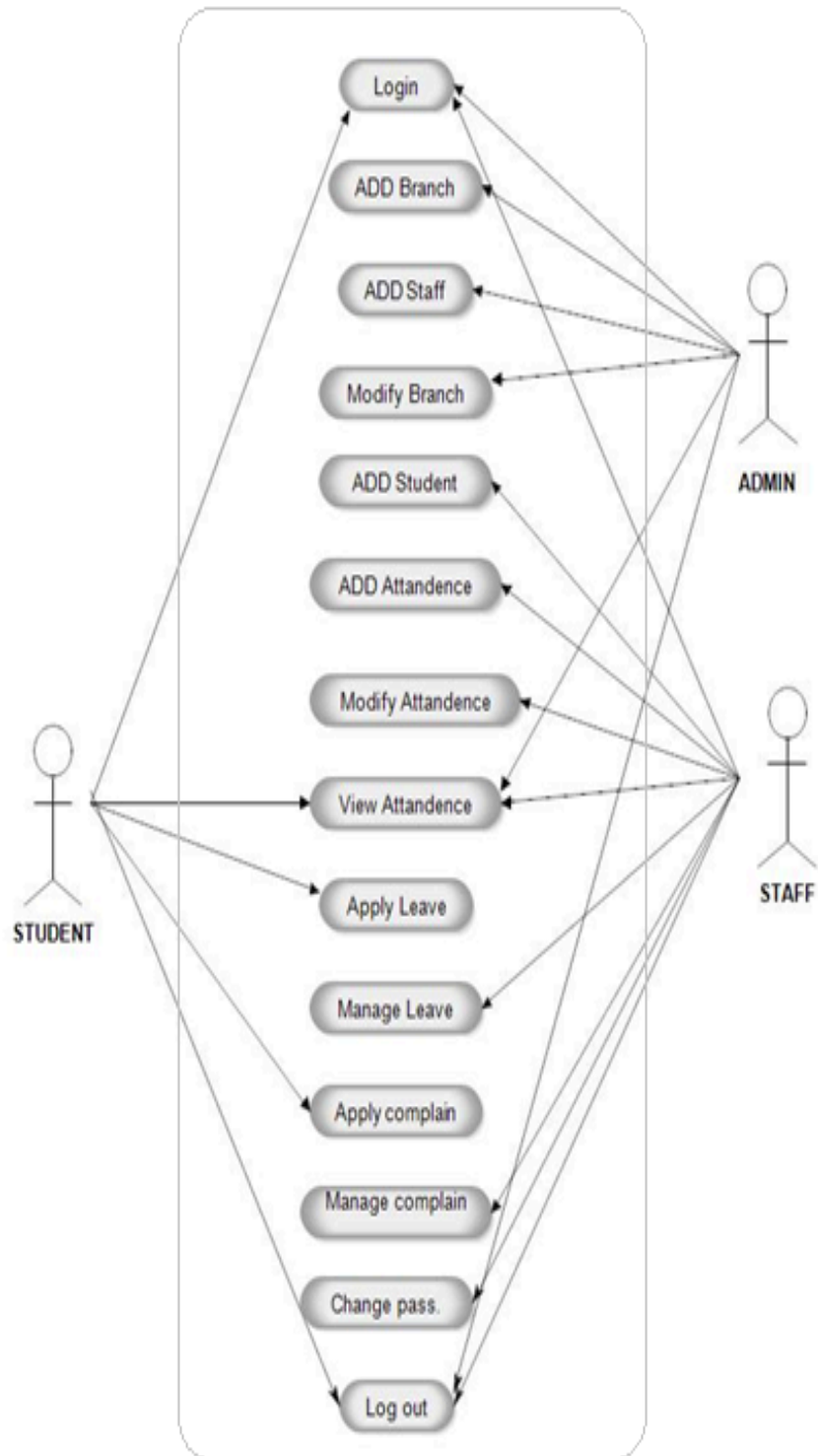
Use Cases:

- Login/Logout
- Add/Update/Delete Student
- Manage Faculty
- Mark/View Attendance
- Generate Reports
- Enter/View Marks
- Pay/View Fees
- Create Timetable

Example Use Case Interactions:

- Admin can manage student/faculty records, fees, timetable, and system configuration.
- Faculty can mark attendance, enter marks, and view timetables.
- Students can view attendance, marks, timetables, and fee status.

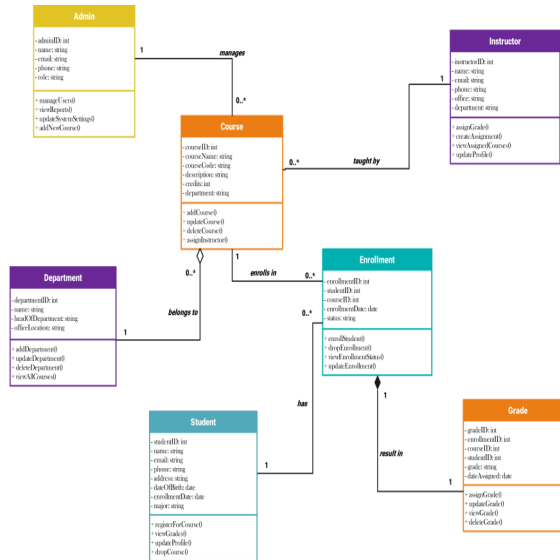
Use case Diagram



UML Diagrams (Unified Modeling Language)

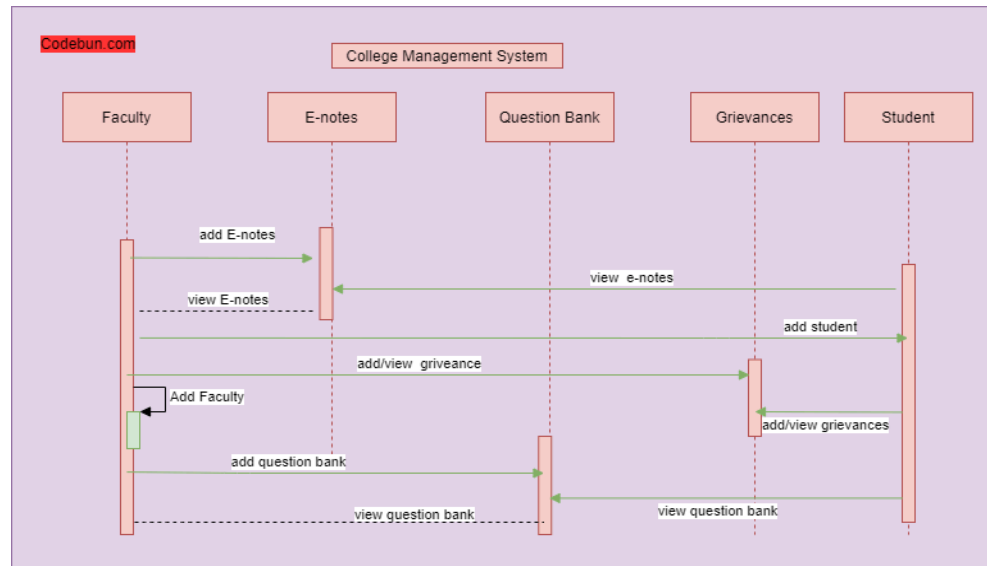
1. Class Diagram:

- Depicts system classes and relationships.
- Classes: Student, Faculty, Admin, Exam, Attendance, Fees, Timetable.



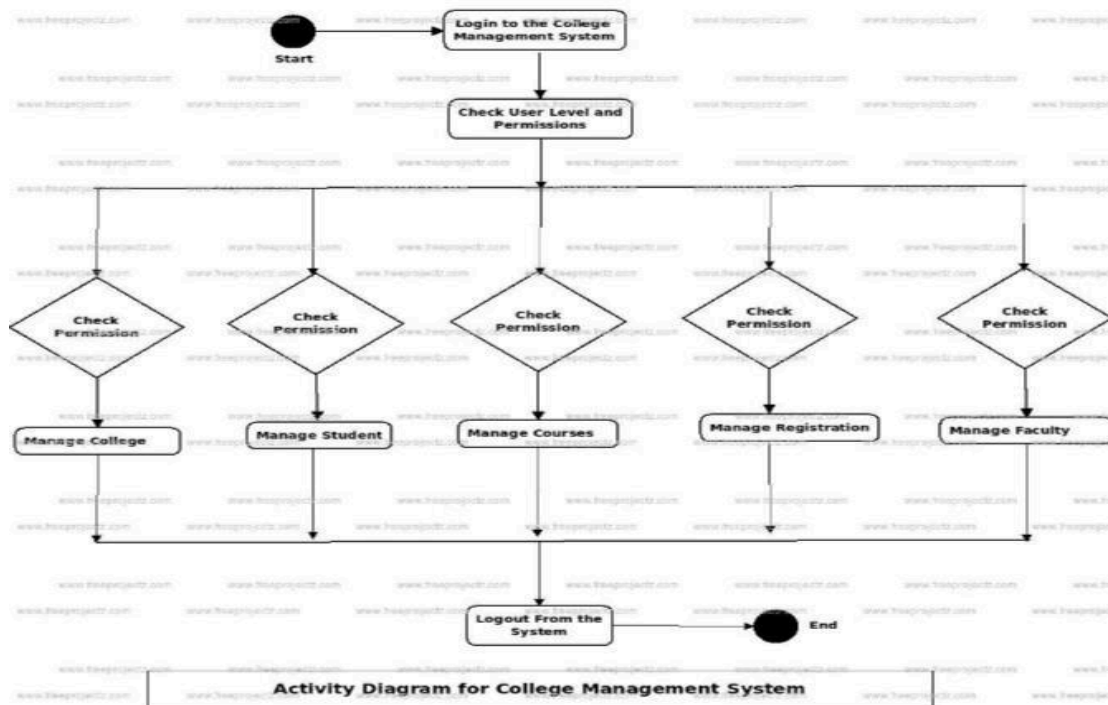
2. Sequence Diagram:

- Shows how objects interact in a time sequence.
- Example: Student login -> System authenticates -> Student dashboard loads -> Student selects module -> Data retrieved/displayed.



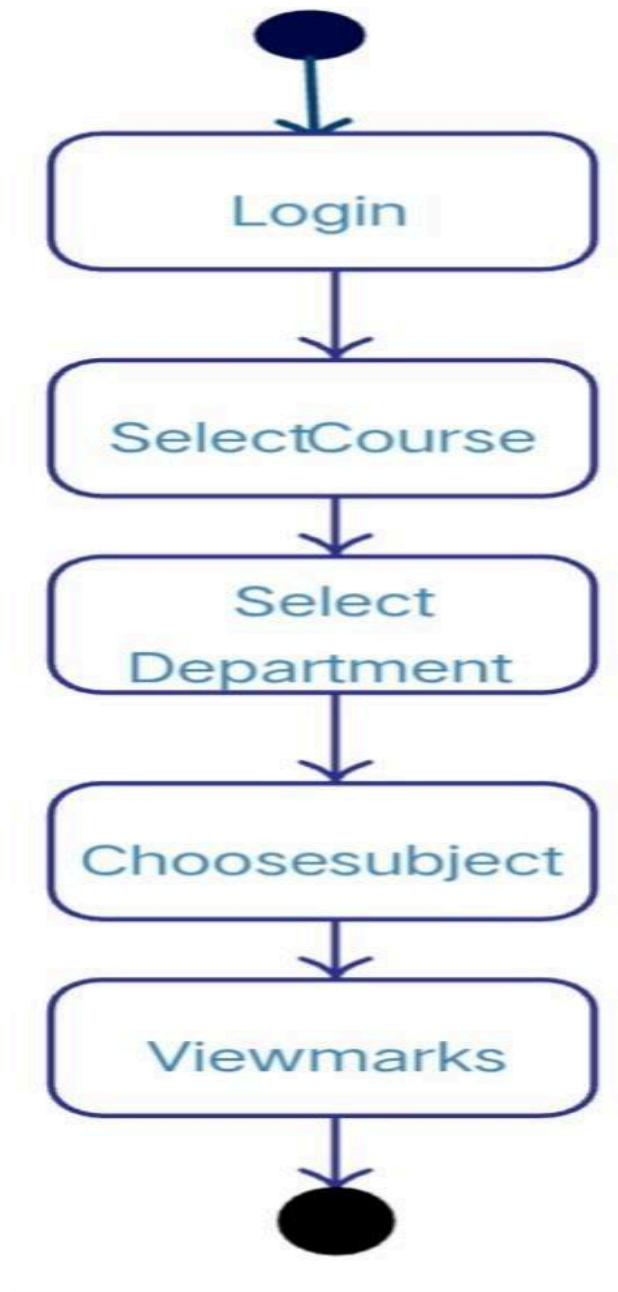
3. Activity Diagram:

- Represents workflows of various modules.
- Example: “Fee Payment” – Start -> Select Payment Option -> Enter Amount -> Confirm Payment -> Generate Receipt -> End.



4. State Transition Diagram:

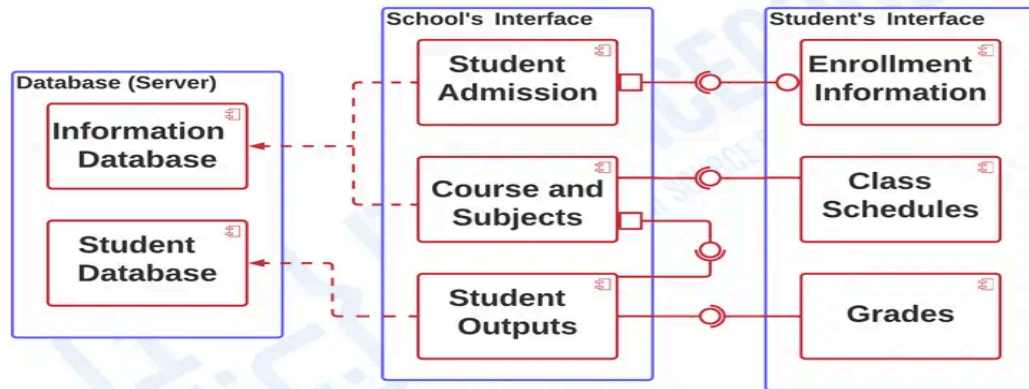
- Shows the different states of a particular object (e.g., Student Status: Registered → Active → Graduated → Archived).



5. Component Diagram:

- Illustrates the software components (Login Module, Attendance Module, etc.) and their interconnections.

COLLEGE MANAGEMENT SYSTEM



COMPONENT DIAGRAM

Input Screen & Output Screen

- MAIN FORM



- STUDENTS LOGIN



- DASHBOARD PAGE



- UPDATE PROFILE



- GALLERY



- FESS DETAILS

localhost:8080/colg%20management%20project/colg%20management%20project/fessdetails.php

Welcome to Kamla Nehru Mahavidyalaya

Dashboard Profile Gallery Fees & Eligibility Leaving Logout

Fees & Eligibility Criteria

COURSE	FEES FOR OPEN CATEGORY	ELIGIBILITY
BSC	₹ 3,379	10+2
BCA	₹20,879	10+2
BCCA	₹23,000	10+2
BCOM	₹8,542	10+2
BA	₹2,581	10+2

Kamla Nehru colg management pr...

Show all

- LEAVING PAGE

localhost:3000/leave-management/leave.php

Welcome to Kanya Nehru Mahavidyalaya

DashboardProfileGalleryFees & EligibilityLeavingLogout

Leaving Page

Leaving Application

Subject

Enter subject

Message

Enter message

Send message

Leaving Data

Sl No.	Date	Name	Subject	Message	Status
1	2024-06-17	shreyashdidi	Geography Forest	sd	pending

Kanya Nehru

leave management p...

Show all

Database Design

(Tables, Attributes, Relationships)

14. Database Design (Tables, Attributes, Relationships)

Database design ensures that data is structured logically and efficiently to support all system functionalities. The College Management System uses a relational database model where tables are related through keys.

14.1 Database Tables and Their Key Attributes

1. Student Table

- * Student_ID (Primary Key)
- * Name
- * Gender
- * Date_of_Birth
- * Contact_No
- * Email
- * Address
- * Course_ID (Foreign Key)

2. Faculty Table

- * Faculty_ID (Primary Key)
- * Name
- * Department
- * Email
- * Contact_No

3. Course Table

- * Course_ID (Primary Key)
- * Course_Name
- * Duration
- * Department

4. Subject Table

- * Subject_ID (Primary Key)
- * Subject_Name
- * Course_ID (Foreign Key)
- * Faculty_ID (Foreign Key)

5. Attendance Table

- * Attendance_ID (Primary Key)
- * Student_ID (Foreign Key)
- * Subject_ID (Foreign Key)
- * Date
- * Status (Present/Absent)

6. Exam Table

- * Exam_ID (Primary Key)
- * Student_ID (Foreign Key)
- * Subject_ID (Foreign Key)
- * Exam_Date

- * Marks_Obtained

7. Fees Table

- * Fee_ID (Primary Key)

- * Student_ID (Foreign Key)

- * Payment_Date

- * Amount

- * Payment_Mode

- * Status (Paid/Pending)

8. Timetable Table

- * Timetable_ID (Primary Key)

- * Course_ID (Foreign Key)

- * Subject_ID (Foreign Key)

- * Day

- * Time_Slot

14.2 Relationships Between Tables

- * A Course can have many Students and many Subjects.

- * A Student can have multiple Attendance, Exam, and Fees records.

- * A Faculty may handle multiple Subjects.

- * A Subject is linked to both Course and Faculty.

- * The Timetable links Course, Subject, and Faculty based on schedules.

Module Description

15. Module Description

This section describes each major functional module of the system.

15.1 Admin Module

Functions:

- Add, update, delete student and faculty records.
- Assign subjects to faculty.
- Manage course and subject data.
- View reports (attendance, exam, fees).
- Manage user access and credentials.

Features:

- Full control of the system.
- Real-time data visibility.
- Secure access to sensitive data.

15.2 Student Module

Functions:

- Login and view personal profile.
- Check attendance records.
- View exam scores and grade reports.
- Access timetable.
- Monitor fee payment status.

Features:

- Personalized dashboard.
- Easy access to academic information.
- Download reports and receipts.

15.3 Faculty Module

Functions:

- Manage student attendance.
- Enter and update exam scores.
- View class and student lists.
- Access assigned timetable.

Features:

- Streamlined interface for academic tasks.
- Real-time data entry and access.

15.4 Attendance Module

Functions:

- Mark daily attendance for each subject.
- Update/edit attendance records.
- Generate monthly or subject-wise reports.

Features:

- Supports subject-wise tracking.
- Exportable reports for admin and faculty.

15.5 Exam Module

Functions:

- Enter exam marks for students.
- Generate mark sheets and grade summaries.
- Generate result reports.

Features:

- Secure data entry and storage.
- Report generation by course/subject.
-

15.6 Fees Module

Functions:

- Record and update fee payments.
- Generate payment receipts.
- View pending dues.

- Notify students of fee status.

Features:

- Tracks payment history.
- Dashboard alerts for due payments.

15.7 Timetable Module

Functions:

- Create and manage course-wise timetables.
- Allocate faculty and subjects to specific time slots.
- View weekly or daily schedules.

Features:

- Dynamic timetable generation.
- Prevents scheduling conflicts.

User Interface Design and Wireframes

User Interface (UI) design focuses on the visual layout and user interaction with the system. A well-designed UI ensures ease of navigation, clarity of options, and accessibility for all user types—Admin, Faculty, and Students.

16.1 Design Principles Used

- Consistency: UI components like buttons, forms, and labels are standardized.
- Simplicity: Clean layout with minimal distractions.
- Feedback: Messages for successful operations and errors (e.g., "Login Successful", "Invalid Password").
- Accessibility: High contrast, readable fonts, and structured forms.

16.2 Key Interface Screens

1. Login Screen

- Fields: Username, Password
- Buttons: Login, Reset
- User Type Selection: Admin / Faculty / Student

2. Admin Dashboard

- Navigation: Manage Students, Manage Faculty, Manage Courses, Timetable, Reports
- Widgets: Recent activity, upcoming events

3. Student Profile Screen

- Tabs: Personal Info, Attendance, Results, Timetable, Fee Status

4. Faculty Panel

- Tabs: Mark Attendance, Enter Marks, View Timetable, Student Lists

5. Timetable View

- Weekly grid format (Days vs. Time Slots)
- Filter by Course and Faculty

6. Fee Payment Interface

- Fee Summary Table
- Payment Status: Paid / Pending
- Receipt Generation Button

7. Reports Module

- Drop-down filters: Course, Semester, Date Range
- Export Options: PDF, Excel

16.3 Wireframe Overview

Since visual wireframes cannot be included directly:

- Each screen follows a header-menu-content-footer layout.
- Left-side panel for menu/navigation.
- Main content area with forms, tables, or dashboards.
- Right-side panels show quick info (notifications or messages).

Implementation Strategy

This phase outlines how the designed system is practically developed, deployed, and managed.

17.1 Development Environment

- Frontend: Visual Basic
- Backend: Oracle
- OS: Windows
- IDE: Visual Studio or VB6 IDE
- Database Tool: Oracle SQL Developer

17.2 Implementation Steps

1. Requirement Finalization

- Interacted with stakeholders (faculty and students).
- Documented functional and non-functional requirements.

2. Environment Setup

- Installed development and database tools.
- Created necessary schema in Oracle.

3. Database Design Implementation

- Created tables, relationships, and constraints.

4. Module-Wise Development

- Developed and tested each module independently (Student, Faculty, Attendance, etc.)

5. User Interface Integration

- Merged module UIs with respective backend functions.

6. Testing

- Conducted thorough unit and integration testing (discussed in detail in the next section).

7. User Acceptance Testing (UAT)

- Took feedback from sample users (admin, faculty, students).

8. Deployment

- Installed on the college system with access rights defined.

9. Training and Documentation

- Provided brief user manual and conducted a training session for admin staff.

Testing & Validation Checks

Testing

Validations are nothing but the security measures taken at the time of execution of any program. It is necessary for the analyst to take validation in his project as it provides more accuracy and systematic flow to the project.

Validation not only stops input of false data but also provides information in the form of message to the user to correct his mistake. Hence it plays a role of a guide during input of data.

Validation put its control over the data in both characters as well as integer data type. Whenever wrong data or invalid data is stored by the user it fires the message immediately and warns the user to input correct data type.

Validations are classified into two types

1. Validating input transaction:

Validating input data is largely done through software which is the programmer's responsibility but it is important that system analyst must know what common problem might invalidate a transaction. Business committed to quality will include validation checks as a part of their routine software

Three main problems occur with input transaction.

1. Submitting the wrong data to system,
2. Submission the data by an unauthorized person or
3. Asking the system to perform an unacceptable function.

2. Validating input data:

It is essential that the input data themselves along with the transactions requested are valid. Several texts can be incorporated into software to ensure this

validity. We consider many possible ways to validate input and they are as follows:

1. Test for missing data
2. Test for correct field length
3. Test for class or composition
4. Test for range or reasonableness
5. Test for comparison with stored data
6. Test for invalid values

System Security Measures

Security is an important term of any project (system) development.

Security provides a protection to the computer, database, and project from unauthorized user. Security of computer facilities stored data and the information generated is a part of successful conversion. Recognition of the need for security is a natural outgrowth of the belief that information is a key organizational resource. It is useful to think of security of the project data and information on an imaginary continuum from totally secure to totally open. Security is the responsibility of all those who come into context with the system. Security has three interrelated aspects i.e. physical, logical and behavioral.

When you run this software, very first form which you find is a password frame consisting of two textboxes for username and password. As soon as frame open and ask for password from the user, user enters the correct password and will move on the next form otherwise move will not be allowed until and unless user types the correct password.

**IMPLEMENTATION, EVALUATION
AND MAINTENANCE.**

- **Implementation**

In this last phase of system development, we have implemented the information system. This phase involves training users to handle the system some training is done by users but oversight of training is the responsibility of the system-analyst. In addition the analyst needs to plan for a smooth conversion from the old system to the new one. This process includes converting files from old formats to zero ones or building a database, installing equipment and bringing the new system into production. It is primary way user training side preparation file conversion when system is link to terminal or remote sides the telecommunication network is also include after implementation. After installation phase is complied and user staff is adjusted and created by system evaluation and maintenance.

- **Evaluation**

Evaluation is shown as part of this final phase of the System Development Life Cycle Mostly for the sake of discussion. Actually, evaluation takes place during every phase. A key criterion that must be satisfied is whether the intended user are indeed using a system throughout at the time of development

We have been evaluating the evolving information system and networks to give feedback for our essential improvement. In recognition the evaluation of information system and network is important. We have taken many evaluation techniques devised. These techniques include cost-benefit analysis models that attempt to estimate the values of a decision based on the effects of revised information using information theory, simulation, user evaluation that

emphasize implementation problem and user evaluation involvement and information system utility approaches that examine the properties of information.

Each type of evaluation serves a different purpose and has invented drawbacks. The revised decision evaluation approach present difficulty because all variables involved within the design, development and information of the project can not be calculated or qualified. The user involvement approach to evaluation can be more comprehensive than others if it is expanded and systematically applied.

- **Maintenance**

After the system is installed it must be maintained. Means the computer programs must be modified and kept up-to-date. Estimates of the time spent by departments on maintenance have ranges from 40-60% of time spent developing projects. Very little time remains for new system development. Maintenance is performed for two reasons. The first of these is to correct software errors. No matters how thoroughly the system is tested, bugs or errors deep into the computer programs. The other reason for performing system maintenance is to enhance the software's capabilities in response to changing organizational needs. Generally involving one of the following two situation

1. Users often request additional features after they become familiar with the computer system and its capabilities
2. Hardware and software are changing at an accelerated phase.

The total cast of maintenance is likely to exceed that of system development. At certain point it becomes more feasible to perform a new information system.

Maintenance is an outgoing process over the life cycle of an information system.

After the system is installed, maintenance usually takes the form of connecting previously undetected program errors. Once these are connected the system approaches a steady state, providing dependable service to its users.

Conclusion

The project entitled as College Management System is the system that deals with the issues related to a particular institution. This project is successfully implemented with all the features mentioned in system requirements specification.

The project entitled as College Management System is the system that deals with the issues related to a particular institution.

- This project is successfully implemented with all the features mentioned in system requirements specification.
- The application provides appropriate information to users according to the chosen service.
- The project is designed keeping in view the day to day problems faced by a college.
- Deployment of our application will certainly help the college to reduce unnecessary wastage of time in personally going to each department for some information.

Awareness and right information about any college is essential for both the development of student as well as faculty. So this serves the right purpose in achieving the desired requirements of both the communities.

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